

Adaptive Lane Keeping with Comparative Yaw Stability Control: Active Front Steering versus Torque Vectoring

by

Ramisha Anjum
Jaiden Dahlke

Abstract

Automated lane keeping becomes increasingly challenging as vehicle speed and path curvature increase, since accurate path tracking must be achieved without inducing excessive sideslip or exceeding the available tire forces. This work investigates how two different lane keeping control approaches, Active Front Steering (AFS) and Torque Vectoring (TV), address this tradeoff. AFS supplements the nominal steering command through additional front wheel steering, whereas TV generates a corrective yaw moment by generating a difference in left and right wheel torques.

The two approaches are first developed and evaluated using a linear bicycle model and are subsequently tested on a CarSim E-Class SUV model. Their performance is compared in terms of cross-track error, yaw response, sideslip angle, lateral acceleration, and actuator effort. The comparison is made over two different maneuver scenarios. The first maneuver tracks a curved road containing a 100 m constant radius segment at a vehicle speed of 60 km/h, and the second is a double lane change at 70 km/h; both with a road friction coefficient of $\mu = 0.90$.

AFS and TV controllers both substantially improve the tracking performance compared to the baseline controller. However, the relative performance depends on the nature of the maneuver. For example, for the curved-road maneuver, the cross-track RMS of the CarSim model reduced from 0.354 m in the baseline case to 0.107 m with AFS and further to 0.066 m with TV. Similarly, for the double lane change, the baseline controller achieves a cross-track RMS of 0.258 m, while AFS achieves the lowest RMS error at 0.096 m and TV achieves 0.136 m.

The observed tradeoff is that TV generally provides strong yaw stability and improved tracking in maneuvers involving sustained cornering, at the expense of greater sideslip and control effort. AFS produces a more moderate corrective response and performs particularly well during transient maneuvers. Hence, neither strategy is universally the better. The respective control approaches depend on the maneuver and the relative importance placed on tracking accuracy and lateral stability.

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I. Introduction

One of the major objectives of automated and highly assisted vehicles is to generate a steering command which follows a lane's center line while remaining dynamically stable. For low to moderate vehicle speeds, the steering input generally produces an acceptable yaw response; hence stability is typically not a concern in this regime. However, for high speeds or sharp lane curvature, the question of stability becomes more complicated. Factors such as tire-force limitations, sideslip, and nonlinear vehicle dynamics can cause a nominal lane-keeping controller to fail.

This interplay creates an important control problem. An appropriate controller must minimize lateral displacement and heading error while also limiting excessive sideslip and avoiding demands for tire forces beyond those available from the road. Should a controller only focus on path position then the resulting transient vehicle motion may become unstable. Conversely, a controller which focuses on vehicle stability too heavily may sacrifice lane-tracking performance. Thus, an effective lane-keeping controller must balance path-following performance with the dynamics limitations of the vehicle.

There are two common methods for modifying the lateral response of a vehicle. Active Front Steering (AFS) adds a corrective front-wheel steering angle to the nominal steering request, while Torque Vectoring (TV) commands different torques between the left and right wheels to directly generate a yaw moment. TV is particularly relevant to electric vehicles where the wheel torques are controlled by independent motors.

This fundamental difference in control strategy raises the central question investigated in this report:

Q: How do steering based AFS controllers and direct yaw moment based TV controllers differ in their ability to improve lane-keeping while preventing adverse effects causing instability?

To answer this question, we implement the following three controllers:

1. **Baseline:** using only the nominal path-following steering command;
2. **AFS:** a constrained optimization-based controller which adds a steering correction;
3. **TV:** the nominal steering is retained and a corrective direct yaw moment is added;

The AFS and TV controllers are both implemented as augmentations of the nominal baseline controller. Hence, the total steering command is

$$\delta_f = \delta_{\text{base}} + \delta_{\text{add}}, \quad (1)$$

where δ_{base} is the baseline controller's steering command and δ_{add} is the supplementary steering provided

by AFS. The outputs of the three controllers are summarized below:

$$\text{Baseline: } \delta_{\text{add}} = 0, \quad M_z = 0, \quad (2)$$

$$\text{AFS: } \delta_{\text{add}} = \delta_{\text{AFS}}, \quad M_z = 0, \quad (3)$$

$$\text{TV: } \delta_{\text{add}} = 0, \quad M_z = M_{z,\text{TV}}. \quad (4)$$

Each controller is implemented on two vehicle models. The first model is a simple linear planar bicycle model, which allows for simpler analysis and greater transparency of the controllers effects. The second is a higher-fidelity CarSim E-Class SUV model. The CarSim plant model provides a far richer representation of the vehicle dynamics, with the drawback of lesser mathematical interpretability.

We test the controllers across both vehicle models with two different maneuver scenarios. The first is a curve contains a constant radius section of 100 m traversed at a speed of 60 km/h with friction coefficient $\mu = 0.90$. This maneuver tests the controllers steady-state tracking, and steady yaw response. The second is a double lane change at 70 km/h with $\mu = 0.90$. In this maneuver, the path moves laterally by 3.5 m and then returns to the original lateral position. The double lane change maneuver causes a rapid curvature reversal, hence it tests transient yaw behavior, overshoot, settling, and sideslip performance. Together, these maneuvers will allow us to assess the controllers performance under a variety of metrics.

Section II reviews previous work on active front steering, torque vectoring and direct yaw-moment control, the coordination of steering and yaw-moment actuation, and their application to combined path-tracking and vehicle-stability control. **Section III** discusses the vehicle models, path-following error, baseline controller along with the AFS and TV formulation, reference signal generation, test scenarios, and measured performance metrics. **Section IV** compares the baseline, AFS, and TV controllers for both maneuvers on both vehicle models, with particular emphasis on the relationship between tracking, stability, and actuator effort. To conclude, **Section V** summarizes the main conclusions, discusses limitations, and future directions of work.

II. Literature Review

Path tracking and vehicle stability are closely related, especially during high-speed maneuvers or when the tire-road friction is limited. A vehicle may remain close to the desired path while developing an undesirable yaw response or excessive sideslip. On the other hand, a controller that focuses mainly on stabilizing yaw motion can change the path-following response. For this reason, both tracking and stability need to be considered when evaluating active vehicle-control systems.

Two commonly studied approaches are Active Front Steering (AFS) and Torque Vectoring (TV). AFS changes the front-wheel steering angle and therefore modifies the front-tire slip angle and lateral force. TV uses unequal longitudinal tire forces on the left and right sides of the vehicle to create a corrective yaw moment. Although both methods can be used to influence yaw motion, they act on the vehicle in different ways. This difference is important when comparing their effects on path tracking, sideslip, lateral acceleration, and actuator effort.

A. AFS for Yaw Stability

AFS has been studied for many years as a way of improving vehicle handling without relying on braking intervention. Instead of replacing the driver's or path-following steering command, AFS adds a corrective front-wheel steering angle. This changes the front-tire lateral force and, as a result, changes both lateral motion and yaw response.

Zheng and Anwar [1] developed an AFS controller for a steer-by-wire vehicle using yaw-rate and front-steering feedback. Their controller also used gain scheduling with vehicle speed so that yaw damping could be improved over a wider operating range. One useful aspect of this paper is that the work was experimentally validated, rather than being limited to simulation. During lane-change tests on packed snow, the controller significantly reduced yaw-rate overshoot. At 64 km/h, the reported overshoot was reduced from about 12 deg/s to about 5 deg/s. At 96 km/h, the uncontrolled vehicle became unstable, while the AFS-controlled vehicle remained stable. This work clearly showed that relatively small steering corrections can have a large effect on vehicle yaw behavior.

A later study by Li et al. [2] applied nonlinear model predictive control (NMPC) to AFS. Their controller used nonlinear vehicle and tire models, including Pacejka tire behavior, to predict the future response of the vehicle and calculate an additional front steering angle. The method was tested using MATLAB/Simulink and CarSim in sinusoidal-steering and double-lane-change maneuvers. The results showed improved yaw stability compared with linear predictive approaches, particularly when the vehicle operated closer to the handling limits. At the same time, the paper also illustrates an important issue: improving yaw-rate tracking does not necessarily mean that sideslip is minimized. A transient sideslip response was still present because sideslip was not treated as an independent control objective.

These studies are useful for the present project for two reasons. First, they show that AFS can improve yaw behavior without applying a direct yaw moment. Second, they show why predictive control is attractive for steering-based stability control. A predictive controller can consider future vehicle motion together with steering limits and steering-rate constraints. However, the effectiveness of AFS still depends on the amount of lateral force that the front tires can generate, which becomes more important near the tire friction limit.

B. Torque Vectoring and Direct Yaw-Moment Control

Torque Vectoring uses a different physical mechanism from AFS. By changing the longitudinal force on the left and right wheels, a yaw moment can be generated directly. This is particularly attractive for electric vehicles with independently controlled motors because wheel torque can be changed quickly and accurately.

Asperti et al. [7] provide a recent review of state-of-the-art TV control. They describe a typical TV system as having three main stages: reference generation, high-level yaw-moment control, and low-level torque allocation. Their review covers a wide range of approaches, including PID, optimal control, sliding-mode control, MPC, and more recent methods. An important point from their review is that TV performance is not determined only by the high-level controller. Reference generation, state estimation, tire friction, actuator limits, and wheel-torque allocation all affect the final vehicle response.

De Novellis et al. [3] compared several feedback controllers for TV in fully electric vehicles. Their study included conventional PID, adaptive PID, and second-order sliding-mode controllers together with a feedforward yaw-moment contribution. The PID-based controllers produced good steady-state and transient performance, while the sliding-mode controllers showed stronger robustness to changes in vehicle parameters. The authors also investigated sideslip control and showed that it becomes especially important when the tire road friction coefficient is not estimated accurately. This is relevant because a yaw-rate target that appears reasonable for one friction condition may no longer be physically achievable when the available road friction is lower.

The relationship between yaw rate and sideslip was examined in more detail by Lu et al. [4]. Their work considered concurrent yaw-rate and sideslip control using torque vectoring on a four-wheel-drive electric vehicle. Nonlinear phase-plane analysis, simulation, and experiments were used to study operation close to the cornering limit. The authors showed that yaw-rate tracking alone is not enough to describe lateral stability. Including sideslip control can enlarge the region in which the vehicle remains stable and can prevent the response from diverging when the requested yaw behavior becomes too aggressive for the available tire friction.

Lenzo et al. [5] discussed another important limitation of direct yaw-moment control. With only one direct yaw-moment input, yaw rate and sideslip cannot in general be controlled independently. Instead of trying to force both variables to arbitrary references, they proposed modifying the yaw-rate reference when sideslip becomes large. Under normal conditions, the controller follows the handling-oriented yaw-rate target. As the sideslip angle approaches a specified threshold, the yaw-rate reference is shifted toward a more stability-oriented value. Their results, which included nonlinear simulation and experimental validation, showed that this approach can keep sideslip within acceptable limits while still maintaining useful yaw-rate control. This idea is particularly important when evaluating TV because the smallest yaw-rate error is not always the safest response.

Jneid and Harth [6] presented a more recent integrated TV controller for an all wheel-motor electric vehicle. Their controller used yaw-rate and sideslip errors and was organized into high-, intermediate-, and low-level control layers. The high-level controller generated the required corrective yaw moment, while the lower layers distributed the demand between the axles and individual wheels. Double-lane-change simulations showed improved heading and stability compared with the uncontrolled vehicle. Their work is also useful because it highlights the additional steps required to move from a desired yaw moment to an actual TV system. Wheel torque, wheel slip, regenerative braking, and torque allocation all need to be considered.

Overall, TV provides strong yaw-control authority because the corrective moment acts directly in the vehicle yaw equation. However, this does not mean that an arbitrarily large yaw moment can be used. The available moment is limited by the motors or brakes and by the friction available at the tires. The reviewed studies also show that yaw-rate tracking should be examined together with sideslip and tire utilization, especially during aggressive maneuvers.

C. Coordination of Steering and Yaw-Moment Actuation

Since AFS and yaw-moment control act through different mechanisms, several studies have investigated using them together. This is useful because the strengths of one actuator can partly compensate for the limitations of the other.

Bianchi et al. [8] studied an integrated controller using AFS together with rear torque vectoring. Their approach used adaptive feedback linearization to account for uncertainties in vehicle mass, yaw inertia, tire stiffness, and road friction. The results showed improved reference tracking even when the vehicle parameters differed from their nominal values. The paper is directly related to the present project because it treats steering and torque vectoring as two different control inputs acting on the same lateral vehicle dynamics.

Ahmadian et al. [9] also coordinated AFS and direct yaw-moment control, but used a model-reference adaptive control framework. Their controller had two layers. The upper layer generated an additive steering command and a corrective yaw moment, while the lower layer converted the yaw moment into rear-wheel braking torque. A stability index was used to determine how the actuators should be used under different conditions. High-speed lane-change simulations showed improved yaw-rate tracking together with lower sideslip and lateral acceleration. The study also considered changes in vehicle mass and tire-road friction, which makes it useful for understanding how actuator coordination can improve robustness.

Research combining AFS with differential braking is also relevant, even though differential braking is not identical to electric torque vectoring. Both methods produce an additional yaw moment by creating unequal longitudinal tire forces on the left and right sides of the vehicle.

Di Cairano et al. [10] coordinated AFS and differential braking using model predictive control. Their vehicle model was written in terms of tire slip angles, and a piecewise affine representation was used to approximate tire behavior as the tires moved from the linear region toward saturation. The authors pointed out an important actuator trade-off. Differential braking provides strong stability intervention but also affects the longitudinal motion of the vehicle, while AFS is less intrusive but has lower stabilizing authority. Experimental tests on low-friction surfaces showed that MPC could coordinate both actuators while respecting their limitations.

Jalali et al. [11] used another MPC formulation to coordinate active steering and differential braking. Their controller included a soft constraint related to vehicle sideslip so that the vehicle could remain within an acceptable stability region while maintaining feasibility of the optimization problem. The controller could be configured to use steering alone, braking alone, or both. The method was evaluated in high-fidelity CarSim simulations and on an instrumented test vehicle. Their results showed that predictive control could intervene before the vehicle reached excessive sideslip, and the real-time optimization was completed within the controller sample time.

These coordinated-control studies show that AFS and direct yaw-moment actuation are complementary rather than identical. AFS changes front lateral force through steering, whereas yaw-moment control acts more directly on vehicle rotation. Using both actuators can improve overall performance, but it also makes it harder to see the individual strengths and disadvantages of each actuator. For that reason, com-

paring them separately under the same operating conditions is still useful.

D. Path Tracking and Yaw Stability in Automated Vehicles

For automated vehicles, path tracking and yaw stability cannot always be treated as separate problems. As lateral acceleration increases, the path-following controller may request a steering response that the tires cannot produce accurately. At the same time, a yaw-stability controller can change the vehicle motion enough to affect the tracking error.

Chatzikomis et al. [12] studied this interaction by combining steering-based path tracking with TV in an autonomous electric vehicle. They compared integrated controllers, where steering and yaw moment were calculated within the same path-tracking controller, with a multi-layer architecture in which the TV controller operated separately from the steering controller. Their obstacle-avoidance simulations showed that TV could improve path tracking, but they also found an important robustness issue. Integrated controllers performed very well when they were tuned for the actual tire-road friction condition, but some became unstable when the friction was reduced without retuning. The separate TV architecture was generally more consistent across a wider range of friction conditions.

Zhang et al. [13] studied a similar problem using MPC for an over-actuated autonomous electric vehicle. Their controllers used both active steering and torque vectoring and included explicit constraints on yaw rate and sideslip angle. Four controller architectures were compared, including one-level and two-level approaches with different wheel-torque allocation methods. The double-lane-change results were evaluated using path-tracking accuracy, passing velocity, tire utilization, robustness, and computational effort. Their results showed that torque vectoring can improve both path-following and stability performance when the available wheel torques are allocated appropriately.

These papers make an important point for the present project. A controller should not be judged only by cross-track error or only by yaw-rate error. A useful comparison should consider both geometric tracking quantities, such as cross-track and heading error, and dynamic quantities, such as yaw rate, sideslip, lateral acceleration, and actuator demand.

E. Summary and Relation to the Present Project

The literature shows that both AFS and TV can improve vehicle handling, but they do so in different ways. AFS works through the front tires by changing steering angle, while TV creates a direct yaw moment through the longitudinal wheel forces. AFS is generally less intrusive to the longitudinal vehicle motion and works naturally with a steering-based path-following controller. TV provides stronger and more direct yaw authority, but its performance depends on available wheel torque, tire friction, and how the requested yaw moment is distributed between the wheels.

Another recurring point in the literature is the importance of sideslip. Several studies show that accurate yaw-rate tracking does not necessarily mean that the vehicle is operating in the most stable condition. This is why yaw rate, sideslip, path error, and control effort should be examined together.

Many previous studies focus either on developing one particular controller or on coordinating several actuators to obtain the best combined performance. The objective of the present project is slightly

different. Instead of combining AFS and TV, the two actuation methods are compared separately while keeping the rest of the problem as consistent as possible. The same baseline path-following controller, vehicle parameters, reference maneuvers, road-friction conditions, and evaluation metrics are used in each case.

The comparison is also performed at two levels of vehicle-model fidelity. A linear bicycle model is first used for controller development and initial evaluation. The same controllers are then tested with a higher-fidelity CarSim vehicle model. This makes it possible to investigate not only whether AFS or TV improves tracking, but also whether the relative conclusions change when more realistic vehicle dynamics are included. The following section describes the vehicle models, controller formulations, simulation architecture, and test conditions used for this comparison.

III. Modeling and Methodology

A. Overall Study Architecture

We designed the project such that all three controllers used the same reference trajectory, and nominal path-following steering controller. The only quantities that differ between controllers are the supplementary control actions provided by the respective control strategies. This design is important because to fairly compare controllers, we must ensure they are receiving the same data.

Figure 1 shows the high-level shared design of our controllers.

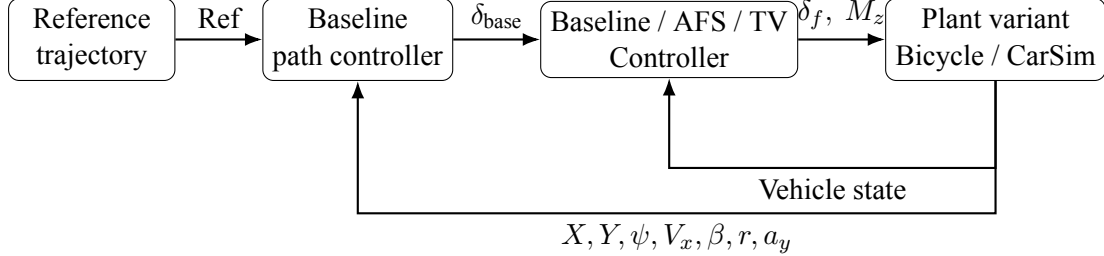


Figure 1: Controller simulation architecture

The controller outputs change depending on the selected controller. They are summarized in Table 1.

Table 1: Control outputs for each controller mode.

Mode	δ_f	M_z
Baseline	δ_{base}	0
AFS	$\delta_{\text{base}} + \delta_{\text{AFS}}$	0
TV	δ_{base}	$M_{z,\text{TV}}$

where δ_{AFS} is the additional steering input provided by the AFS controller and $M_{z,\text{TV}}$ is the additional yaw moment provided by the TV controller.

B. Vehicle Modeling Assumptions

The controller-design model is a planar, two-degree-of-freedom linear bicycle model. The primary assumptions are:

1. the vehicle is represented by one equivalent front wheel and one equivalent rear wheel;
2. longitudinal speed V_x is treated as constant over the lateral model dynamics;
3. roll, pitch, heave, suspension motion, and wheel-hop dynamics are neglected;
4. front and rear lateral tire forces are linear with slip angle in the controller-design model;
5. steering and vehicle angles are sufficiently small for linear lateral-dynamics relationships;
6. aerodynamic lateral forces are neglected; and
7. TV is represented by a net yaw moment M_z acting about the vehicle center of gravity.

These assumptions intentionally produce a transparent model suitable for controller design. They are subsequently relaxed implicitly by replacing the bicycle plant with the higher-fidelity CarSim vehicle.

The parameters used by the MATLAB/Simulink controller-design model are listed in Table 2.

Table 2: E-Class vehicle parameters used in the controller-design model.

Parameter	Symbol	Value
Vehicle mass	m	1919.2 kg
Yaw moment of inertia	I_z	2687.1 kg m ²
CG to front axle	a	1.180 m
CG to rear axle	b	1.770 m
Wheelbase	$L = a + b$	2.950 m
Front axle cornering stiffness	C_f	145466.47 N/rad
Rear axle cornering stiffness	C_r	98929.60 N/rad

C. Planar Bicycle Model

The lateral-dynamics states are vehicle sideslip angle β and yaw rate r :

$$\mathbf{x} = \begin{bmatrix} \beta \\ r \end{bmatrix}. \quad (5)$$

For small sideslip angle,

$$\beta \approx \frac{V_y}{V_x}, \quad (6)$$

where V_y is the lateral velocity at the center of gravity.

The front and rear tire slip-angle relationships are written as

$$\alpha_f = \delta_f - \beta - \frac{ar}{V_x}, \quad (7)$$

$$\alpha_r = -\beta + \frac{br}{V_x}. \quad (8)$$

Within the linear tire region,

$$F_{yf} = C_f \alpha_f, \quad (9)$$

$$F_{yr} = C_r \alpha_r. \quad (10)$$

The lateral force balance is

$$mV_x (\dot{\beta} + r) = F_{yf} + F_{yr}, \quad (11)$$

and the yaw-moment balance about the center of gravity is

$$I_z \dot{r} = aF_{yf} - bF_{yr} + M_z. \quad (12)$$

Substitution of Equations (7)–(10) into Equations (11) and (12) gives

$$\dot{\mathbf{x}} = A(V_x)\mathbf{x} + B_\delta(V_x)\delta_f + B_M M_z, \quad (13)$$

where

$$A(V_x) = \begin{bmatrix} -\frac{C_f + C_r}{mV_x} & -\frac{aC_f - bC_r + mV_x^2}{mV_x^2} \\ -\frac{aC_f - bC_r}{I_z} & -\frac{a^2C_f + b^2C_r}{I_z V_x} \end{bmatrix}, \quad (14)$$

$$B_\delta(V_x) = \begin{bmatrix} \frac{C_f}{mV_x} \\ \frac{aC_f}{I_z} \end{bmatrix}, \quad B_M = \begin{bmatrix} 0 \\ \frac{1}{I_z} \end{bmatrix}. \quad (15)$$

Equation (13) directly demonstrates the fundamental difference between AFS and TV. Steering first modifies the front-tire lateral force and therefore affects both the lateral and yaw equations. The TV command enters the yaw-rate equation directly through B_M .

The global vehicle motion used for path tracking is obtained from

$$\dot{X} = V_x \cos \psi - V_y \sin \psi, \quad (16)$$

$$\dot{Y} = V_x \sin \psi + V_y \cos \psi, \quad (17)$$

$$\dot{\psi} = r, \quad (18)$$

where (X, Y) is the center-of-gravity position in the inertial frame and ψ is vehicle heading.

D. Understeer Characteristic and Steady Yaw Response

The linear understeer coefficient provides useful physical interpretation of the steering-to-yaw relationship:

$$K_{\text{us}} = \frac{m}{L} \left(\frac{b}{C_f} - \frac{a}{C_r} \right). \quad (19)$$

Using the project vehicle parameters gives approximately $K_{\text{us}} = 1.56 \times 10^{-4} \text{ s}^2/\text{m}$. Because $K_{\text{us}} > 0$, the linear model has an understeering characteristic.

The steady yaw rate associated with a front road-wheel angle δ_f can therefore be interpreted as

$$r_{\text{ss}} = \frac{V_x}{L + K_{\text{us}} V_x^2} \delta_f. \quad (20)$$

At 60 km/h, the corresponding linear steady yaw-rate gain is approximately $r_{\text{ss}}/\delta_f \approx 5.57 \text{ s}^{-1}$; at 70 km/h, it is approximately $r_{\text{ss}}/\delta_f \approx 6.46 \text{ s}^{-1}$.

The implementation does not rely only on these scalar equations. The AFS optimization computes the dynamically consistent steady-state sideslip reference directly from the state-space matrices, as discussed later. Nevertheless, Equation (20) is useful for understanding how nominal steering corresponds to a desired yaw response.

E. Path-Tracking Errors

Let $(X_{\text{ref}}, Y_{\text{ref}}, \psi_{\text{ref}})$ represent the position and heading of the reference path. A signed cross-track error is calculated in the direction normal to the reference heading:

$$e_y = -\sin(\psi_{\text{ref}})(X - X_{\text{ref}}) + \cos(\psi_{\text{ref}})(Y - Y_{\text{ref}}). \quad (21)$$

This definition is preferable to simply using $Y - Y_{\text{ref}}$ because it remains meaningful when the reference path rotates during the constant-radius curve.

Heading error is calculated using a wrapped angular difference:

$$e_\psi = \text{atan2} [\sin(\psi_{\text{ref}} - \psi), \cos(\psi_{\text{ref}} - \psi)]. \quad (22)$$

The use of the atan2 form prevents a discontinuity when an angular error crosses $\pm\pi$.

F. Nominal Baseline Steering Controller

The Baseline controller is retained in every test case and represents the nominal lane/path-following function. Its purpose is to produce the primary steering command δ_{base} from reference geometry and tracking errors. The configured feedback gains are $K_y = 0.02$ and $K_\psi = 0.5$, and the steering command is limited to $|\delta_{\text{base}}| \leq 30^\circ$.

At a block-diagram level, the nominal controller can be represented by

$$\delta_{\text{base}} = \text{sat}(\delta_{\text{ff}} + \delta_{\text{fb}}, \pm\delta_{\text{max}}), \quad (23)$$

where δ_{ff} represents the path-geometry contribution and δ_{fb} uses lateral and heading errors. The exact signal implementation is retained unchanged among Baseline, AFS, and TV simulations. Therefore, the purpose of this project is not to compare different path controllers; it is to compare two different methods of augmenting the same path controller.

G. Active Front Steering

G.1 Physical Principle

AFS creates a corrective yaw response by modifying the front steering angle:

$$\delta_f = \delta_{\text{base}} + \delta_{\text{AFS}}. \quad (24)$$

The physical sequence is

$$\delta_{\text{AFS}} \rightarrow \alpha_f \rightarrow F_{yf} \rightarrow aF_{yf} \rightarrow r.$$

Therefore, AFS does not inject a yaw moment independently of the tires. Instead, it changes the front lateral tire force that naturally produces both lateral acceleration and yaw moment.

This coupling is an important feature. It allows AFS to influence the vehicle through the same front-tire mechanism that already performs path steering, but it also means that its yaw authority depends on available front-tire lateral force.

G.2 Discrete Prediction Model

The AFS implementation uses the state vector

$$\mathbf{x}_k = \begin{bmatrix} \beta_k \\ r_k \end{bmatrix}, \quad (25)$$

with

$$\delta_{f,k} = \delta_{\text{base},k} + \delta_{\text{AFS},k}. \quad (26)$$

The continuous bicycle model is discretized using an exact zero-order-hold transformation:

$$\mathbf{x}_{k+1} = A_d \mathbf{x}_k + B_d (\delta_{\text{base},k} + \delta_{\text{AFS},k}). \quad (27)$$

The exact matrix exponential is evaluated as

$$\exp \left(\begin{bmatrix} A & B_\delta \\ 0 & 0 \end{bmatrix} T_s \right) = \begin{bmatrix} A_d & B_d \\ 0 & 1 \end{bmatrix}. \quad (28)$$

The AFS controller sample time is $T_s = 0.02$ s and the prediction horizon is $N_p = 25$. Thus, the controller predicts approximately $T_p = N_p T_s = 0.5$ s into the future.

G.3 Steady-State Sideslip Reference

One important implementation detail is that the AFS controller does *not* force the desired sideslip to zero during steady cornering. A nonzero sideslip angle is physically required for the tires to generate the lateral forces needed for circular motion.

For a specified desired yaw rate r_{ref} , the steady-state condition is

$$\mathbf{0} = A \begin{bmatrix} \beta_{\text{ref}} \\ r_{\text{ref}} \end{bmatrix} + B_{\delta} \delta_{\text{eq}}, \quad (29)$$

where δ_{eq} is the equilibrium steering angle associated with the desired yaw rate.

The first and second state equations can be rearranged as

$$\begin{bmatrix} A_{11} & B_1 \\ A_{21} & B_2 \end{bmatrix} \begin{bmatrix} \beta_{\text{ref}} \\ \delta_{\text{eq}} \end{bmatrix} = - \begin{bmatrix} A_{12} \\ A_{22} \end{bmatrix} r_{\text{ref}}. \quad (30)$$

The resulting coefficient relating sideslip to yaw rate is computed when the MPC matrices are constructed. The reference used by the optimizer is then

$$\mathbf{x}_{\text{ref}} = \begin{bmatrix} \beta_{\text{ref}} \\ r_{\text{ref}} \end{bmatrix}. \quad (31)$$

This makes the AFS objective dynamically consistent with steady vehicle cornering.

G.4 Prediction Matrices

Over the prediction horizon, the future state sequence can be written as

$$\mathbf{X} = \Phi \mathbf{x}_0 + \Gamma \mathbf{U} + \Gamma \Delta_{\text{base}}, \quad (32)$$

where

$$\mathbf{U} = \begin{bmatrix} \delta_{\text{AFS},0} & \delta_{\text{AFS},1} & \cdots & \delta_{\text{AFS},N_p-1} \end{bmatrix}^T. \quad (33)$$

The baseline steering is assumed constant over the current optimization horizon:

$$\Delta_{\text{base}} = \delta_{\text{base}} \mathbf{1}_{N_p}. \quad (34)$$

This approximation reduces complexity while the optimization is repeated at every controller sample.

The prediction matrices are

$$\Phi = \begin{bmatrix} A_d \\ A_d^2 \\ \vdots \\ A_d^{N_p} \end{bmatrix}, \quad (35)$$

and

$$\Gamma = \begin{bmatrix} B_d & 0 & \cdots & 0 \\ A_d B_d & B_d & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ A_d^{N_p-1} B_d & A_d^{N_p-2} B_d & \cdots & B_d \end{bmatrix}. \quad (36)$$

G.5 AFS Cost Function

The AFS controller minimizes a quadratic objective containing three components:

1. state-tracking error;
2. additional-steering magnitude; and
3. change in additional steering.

The finite-horizon objective is

$$\begin{aligned} J = & \sum_{i=1}^{N_p} (\mathbf{x}_{k+i} - \mathbf{x}_{\text{ref},k+i})^T Q (\mathbf{x}_{k+i} - \mathbf{x}_{\text{ref},k+i}) \\ & + \sum_{i=0}^{N_p-1} R \delta_{\text{AFS},k+i}^2 \\ & + \sum_{i=0}^{N_p-1} R_d (\Delta \delta_{\text{AFS},k+i})^2. \end{aligned} \quad (37)$$

The implemented weighting matrices are

$$Q = \begin{bmatrix} 50 & 0 \\ 0 & 300 \end{bmatrix}, \quad R = 5, \quad R_d = 200. \quad (38)$$

Within Q , yaw rate receives greater weight than sideslip. The magnitude and difference penalties prevent unnecessarily large or abrupt steering corrections.

For the stacked problem,

$$Q_{\text{bar}} = I_{N_p} \otimes Q, \quad (39)$$

$$R_{\text{bar}} = R I_{N_p}, \quad (40)$$

$$R_{d,\text{bar}} = R_d I_{N_p}. \quad (41)$$

A difference matrix E is used such that

$$EU + \mathbf{e}_{\text{prev}} = \begin{bmatrix} u_0 - u_{\text{prev}} \\ u_1 - u_0 \\ \vdots \\ u_{N_p-1} - u_{N_p-2} \end{bmatrix}. \quad (42)$$

The resulting quadratic program has the standard form

$$\min_U \quad \frac{1}{2} U^T H U + f^T U, \quad (43)$$

where

$$H = 2 \left(\Gamma^T Q_{\text{bar}} \Gamma + R_{\text{bar}} + E^T R_{d,\text{bar}} E \right). \quad (44)$$

A small numerical regularization term, $10^{-9}I$, is added to H to improve numerical robustness.

G.6 AFS Constraints

The optimizer enforces $|\delta_{\text{AFS}}| \leq 5^\circ$, $|\dot{\delta}_{\text{AFS}}| \leq 250^\circ/\text{s}$, and $|\delta_{\text{base}} + \delta_{\text{AFS}}| \leq 30^\circ$. Because $T_s = 0.02 \text{ s}$, the rate limit corresponds to a maximum change of $250^\circ/\text{s} \times 0.02 \text{ s} = 5^\circ$ per AFS controller sample.

The combined angle bounds are calculated dynamically because the available AFS correction depends on the current nominal steering command:

$$\delta_{\text{AFS},\text{min}} = \max(-\delta_{\text{AFS},\text{max}}, -\delta_{\text{total},\text{max}} - \delta_{\text{base}}), \quad (45)$$

$$\delta_{\text{AFS},\text{max},\text{eff}} = \min(\delta_{\text{AFS},\text{max}}, \delta_{\text{total},\text{max}} - \delta_{\text{base}}). \quad (46)$$

The quadratic program is solved using MATLAB `quadprog` with the `interior-point-convex` algorithm. Only the first optimized steering command is applied:

$$\delta_{\text{AFS},k} = U_0^*. \quad (47)$$

At the next controller sample, the current state is measured again and the entire optimization is repeated. This is the receding-horizon principle.

If a valid optimization solution cannot be obtained, the implementation retains the previous AFS command while clipping it to the currently feasible angle bounds. A non-finite command is replaced by zero. These fallback rules prevent a numerical optimization failure from generating an unbounded steering request.

H. Torque Vectoring / Direct Yaw-Moment Controller

H.1 Physical Principle

Torque vectoring generates yaw moment through a left–right longitudinal force difference. For a track width t_w , the basic relationship can be represented approximately as

$$M_{z,\text{TV}} \approx \frac{t_w}{2} (F_{x,R} - F_{x,L}). \quad (48)$$

Unlike AFS, this yaw-control channel does not require a change in front steering angle. In the present project, wheel-level motor torque distribution is outside the scope. Instead, the dynamically equivalent yaw moment $M_{z,\text{TV}}$ is generated by the Simulink controller and applied to CarSim through its external-yaw-moment interface. Consequently, the study compares *steering actuation* against *direct yaw-moment*

actuation; it does not evaluate a specific electric-motor torque allocator.

H.2 Reference Filtering

The TV architecture is based on a conventional feedforward-plus-PID yaw controller [3]. A first-order filter is applied to the desired yaw-rate signal:

$$\tau_{\text{ref}} \dot{r}_f + r_f = r_{\text{cmd}}, \quad (49)$$

with $\tau_{\text{ref}} = 0.3$ s. The yaw-rate tracking error is

$$e_r = r_f - r. \quad (50)$$

Filtering reduces abrupt changes in requested yaw response and is particularly useful when the path curvature changes rapidly.

H.3 PID Feedback

The parallel PID feedback contribution can be represented as

$$M_{z,\text{FB}}^* = K_p e_r + M_I + K_d \dot{e}_r, \quad (51)$$

where M_I is the integral contribution. The active gains retained in the project are $K_p = 80\,000$ N m s/rad, $K_i = 0.004$ N m/rad, and $K_d = 0.8$ N m s²/rad.

These gains originate from the conventional PID case used as the literature baseline in De Novellis *et al.* [3]. The original operating point associated with the published values was 90 km/h. The complete speed-scheduling maps used in the source study were not available. Therefore, the project retains the published gains as a traceable baseline rather than silently inventing a new gain schedule. Vehicle-speed dependence still enters the reference generation and model-based feedforward calculations.

H.4 Quasi-Static Feedforward

Feedback acts only after a yaw-rate error is created. To reduce the amount of corrective work required from feedback, the TV controller also uses a quasi-static feedforward yaw moment.

For a given nominal steering angle δ_{base} and desired yaw rate r_f , the steady sideslip required by the first state equation is

$$\beta_{\text{ss}} = -\frac{A_{12}r_f + B_1\delta_{\text{base}}}{A_{11}}. \quad (52)$$

The yaw moment required to satisfy the second steady-state equation is then

$$M_{z,\text{FF}} = -I_z (A_{21}\beta_{\text{ss}} + A_{22}r_f + B_2\delta_{\text{base}}). \quad (53)$$

Equivalently, from the steady yaw-moment balance,

$$M_{z,\text{FF}} = bF_{yr,\text{ss}} - aF_{yf,\text{ss}}. \quad (54)$$

The feedforward term therefore compensates the yaw-moment imbalance that would otherwise need to be corrected entirely by feedback.

The unsaturated requested TV moment is

$$M_z^* = M_{z,\text{FF}} + M_{z,\text{FB}}^*. \quad (55)$$

H.5 Yaw-Moment Saturation and Anti-Windup

The available direct yaw-moment authority is limited to $|M_z| \leq 5000 \text{ N m}$.

Thus,

$$M_z = \text{sat}(M_z^*, -5000, 5000). \quad (56)$$

When saturation occurs, the controller cannot realize the moment requested by the PID calculation. Continuing to integrate yaw-rate error under this condition would cause integral windup. Back-calculation anti-windup is therefore used. In terms of the integral moment contribution, a dimensionally convenient representation is

$$\dot{M}_I = K_i e_r + K_{\text{aw}} (M_z - M_z^*), \quad (57)$$

with $\tau_{\text{aw}} = 0.1 \text{ s}$ and $K_{\text{aw}} = 1/\tau_{\text{aw}} = 10 \text{ s}^{-1}$.

The saturation difference therefore drives the integral contribution back toward a value consistent with the moment that can physically be applied.

I. Common Yaw-Rate Reference and Friction Limit

Two closely related yaw-rate references are used in the project and should be distinguished.

For **controller operation**, the nominal steering command defines the desired steady vehicle yaw response. This is interpreted through the vehicle's steering-to-yaw relationship and is clipped according to the available road friction. AFS uses the active reference in its prediction problem, while TV additionally filters the reference according to Equation (49).

For **performance evaluation**, a common path-based yaw-rate target is calculated from the reference curvature:

$$r_{\text{path}} = V_x \kappa_{\text{ref}}. \quad (58)$$

The friction-limited yaw-rate magnitude is approximated by

$$r_{\mu,\text{max}} = \frac{\mu g}{\max(|V_x|, 0.1)}. \quad (59)$$

The reference used for comparison is therefore

$$r_{\text{ref}} = \text{clip}(V_x \kappa_{\text{ref}}, -r_{\mu,\text{max}}, r_{\mu,\text{max}}). \quad (60)$$

At 60 km/h and $\mu = 0.90$, $r_{\mu,\text{max}} \approx 0.530 \text{ rad/s}$; at 70 km/h, $r_{\mu,\text{max}} \approx 0.454 \text{ rad/s}$.

For the 100 m constant-radius curve, $\kappa = 1/R = 0.01 \text{ m}^{-1}$, so the nominal path yaw rate is $r_{\text{path}} = 16.667/100 = 0.1667 \text{ rad/s}$.

This is comfortably below the friction-based yaw-rate limit for the selected dry-road condition. The corresponding kinematic lateral acceleration is $a_y = V_x^2/R = (16.667)^2/100 \approx 2.78 \text{ m/s}^2$.

J. MATLAB/Simulink Implementation

The project is implemented using a modular Simulink architecture. A common vehicle-state bus contains $\{X, Y, \psi, V_x, V_y, r, \beta, a_y\}$, whereas the reference bus contains $\{X_{\text{ref}}, Y_{\text{ref}}, \psi_{\text{ref}}, \kappa_{\text{ref}}, V_{x,\text{ref}}, \mu\}$.

The mutually exclusive controller modes are Baseline (0), AFS (1), and TV (2).

Only one controller variant is active during a simulation. A corresponding plant architecture permits the same controller signal interface to be connected either to the linear bicycle model or to the CarSim adapter.

The AFS model contains discrete states and runs as a fixed-step discrete referenced model. The optimization itself is executed at $T_s = 0.02 \text{ s}$ even though the parent CarSim-linked simulation uses a considerably smaller numerical integration step.

The TV model contains continuous controller states associated with its filter and PID implementation and is therefore solved with a fixed-step continuous solver when referenced from the CarSim harness.

K. CarSim--Simulink Co-Simulation

The bicycle model is useful for control development, but it does not represent all effects present in a real road vehicle. Higher-fidelity validation was therefore performed with CarSim 2025.0 using an E-Class SUV dataset.

Simulink generates the nominal steering and AFS/TV augmentation. CarSim acts as the vehicle plant and returns the measured vehicle states.

The Simulink-to-CarSim import interface is listed in Table 3.

Table 3: Control signals imported from Simulink into CarSim.

Index	CarSim variable	Unit	Mode
1	IMP_STEER_L1	deg	Replace
2	IMP_STEER_R1	deg	Replace
3	IMP_MZ_EXT	N m	Add

The total Simulink steering command is generated in radians and converted to degrees before being sent to both front-wheel steering imports. For the TV case, the external-yaw-moment command is sent directly in N m.

The exported CarSim signals are listed in Table 4.

Table 4: CarSim signals exported to the Simulink vehicle-state interface.

Index	CarSim variable	CarSim unit
1	Xcg_TM	m
2	Ycg_TM	m
3	Yaw	deg
4	Vx	km/h
5	Vy	km/h
6	AVz	deg/s
7	Beta	deg
8	Ay	g
9	Steer_L1	deg
10	Steer_R1	deg

The interface converts these quantities to the common SI-unit vehicle state. Specifically,

$$\psi_{\text{rad}} = \psi_{\text{deg}} \frac{\pi}{180}, \quad (61)$$

$$r_{\text{rad/s}} = r_{\text{deg/s}} \frac{\pi}{180}, \quad (62)$$

$$V_{x,\text{m/s}} = \frac{V_{x,\text{km/h}}}{3.6}, \quad (63)$$

$$V_{y,\text{m/s}} = \frac{V_{y,\text{km/h}}}{3.6}, \quad (64)$$

$$a_{y,\text{m/s}^2} = 9.81 a_{y,g}. \quad (65)$$

This common interface allows the controllers to operate without being rewritten when the plant changes from the bicycle model to CarSim.

The final CarSim simulations use $\Delta t_{\text{CarSim}} = 0.0005$ s. The CarSim output-file interval is 0.025 s, while the Simulink parent model also runs at 0.0005 s.

The referenced controller solver settings are summarized in Table 5.

Table 5: Fixed-step settings used for the final CarSim co-simulations.

Component	Solver	Step size
Parent Simulink model	Fixed step	0.0005 s
CarSim math model	Fixed step	0.0005 s
AFS referenced model	FixedStepDiscrete	0.0005 s
TV referenced model	ode4	0.0005 s

The much smaller plant-integration step relative to the 0.02 s AFS optimization period is intentional. CarSim and the vehicle plant are integrated at high resolution, whereas the steering optimizer updates at a realistic lower control frequency.

L. Test Maneuvers

Two maneuvers were selected, as summarized in Table 6.

Table 6: Final test conditions.

Maneuver	Speed	μ	Geometry	Metric window
Constant-radius curve	60 km/h	0.90	$R = 100$ m	3–10 s
Double lane change	70 km/h	0.90	3.5 m shift	1.54–8.49 s

L.1 Constant-Radius Curve

The curved-lane maneuver begins with straight-line travel at 60 km/h. The curve begins at $t = 2$ s.

The straight-line distance before curve entry is therefore $s_{\text{entry}} = V_x t = 16.667(2) \approx 33.33$ m.

The path then follows a 100 m radius left-hand curve, with CarSim road friction set to $\mu = 0.90$.

The total simulation duration is 10 s. Performance metrics are evaluated from 3 to 10 s so that the complete plots retain the curve-entry transient while the principal numerical metrics focus on the response after the first second of turn entry.

L.2 Double Lane Change

The double lane change is performed at $V_x = 70$ km/h = 19.444 m/s.

The path moves laterally by one lane width, $\Delta Y = 3.5$ m, using approximately 45 m transition segments and a 25 m hold section before returning to the original lane. The complete simulation duration is 12 s.

The evaluation interval is approximately $1.543 \text{ s} \leq t \leq 8.486 \text{ s}$, which corresponds to the active maneuver region rather than including long periods of straight driving before and after the lane changes.

M. Performance Metrics

Tracking, yaw dynamics, stability, and actuator effort are all considered.

For any error signal $e(t)$ evaluated over N samples, RMS error is

$$e_{\text{RMS}} = \sqrt{\frac{1}{N} \sum_{k=1}^N e_k^2}. \quad (66)$$

Peak magnitude is

$$e_{\text{peak}} = \max_k |e_k|. \quad (67)$$

The principal metrics are:

1. cross-track RMS and peak error;
2. heading RMS and peak error;
3. yaw-rate RMS and peak tracking error;

4. peak vehicle sideslip angle;
5. peak lateral acceleration;
6. total front-steering demand;
7. AFS additional-steering magnitude and rate;
8. TV corrective-yaw-moment magnitude; and
9. actuator saturation fraction where applicable.

The intention is not simply to select whichever controller produces the smallest path error. A controller is considered well balanced only when tracking improvement is interpreted together with sideslip, lateral acceleration, and actuator demand.

IV. Simulation Results and Discussions

A. Evaluation Approach

The controllers were first tested using the linear bicycle model and were then evaluated using the higher-fidelity CarSim vehicle model. The bicycle model was useful for checking the controller behavior and comparing the basic effects of Baseline, AFS, and TV under simplified vehicle dynamics. The same controller structure was then connected to CarSim to determine how the results changed when more detailed vehicle dynamics were included. In total, twelve main simulation cases were considered: $2 \text{ vehicle models} \times 3 \text{ controller modes} \times 2 \text{ maneuvers} = 12 \text{ cases}$. The results are discussed using tracking performance, yaw response, sideslip angle, lateral acceleration, and actuator effort. This allows the controllers to be compared based on more than only path error.

B. Co-Simulation Interface Verification

Before comparing the controller results, the CarSim interface was checked to confirm that the steering commands generated in Simulink were being applied correctly in CarSim. The commanded steering was compared with the actual left- and right-front road-wheel angles reported by CarSim. The longitudinal speed was also checked to make sure that all three controller cases were being tested at approximately the same forward speed.

This verification was necessary because steering is handled in radians inside the Simulink controller, while the CarSim steering inputs are defined in degrees. The plots show that the commanded steering is correctly passed to the CarSim model. Small differences between the reported left and right wheel angles are expected because CarSim includes the actual steering geometry of the vehicle. The CarSim speed controller also maintained approximately 16.67 m/s during the curved-lane test and 19.44 m/s during the double lane change. Therefore, the differences discussed in the following sections mainly come from the lateral and yaw controllers rather than from differences in longitudinal speed.

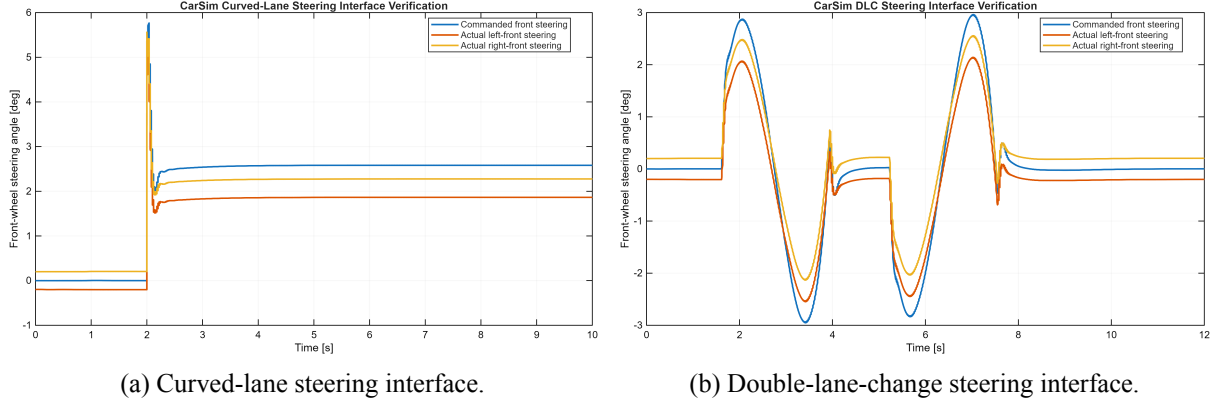


Figure 2: Verification of the Simulink-to-CarSim front-steering interface. The commanded road-wheel steering is delivered to CarSim, while the actual left and right wheel angles reflect the vehicle steering geometry.

C. Constant-Radius Curve: Bicycle-Model Results

The curved-lane test was first carried out using the bicycle model. Since this model uses linear tire forces and does not include effects such as load transfer, suspension motion, or steering compliance, the response is expected to be smoother and more predictable than the CarSim response.

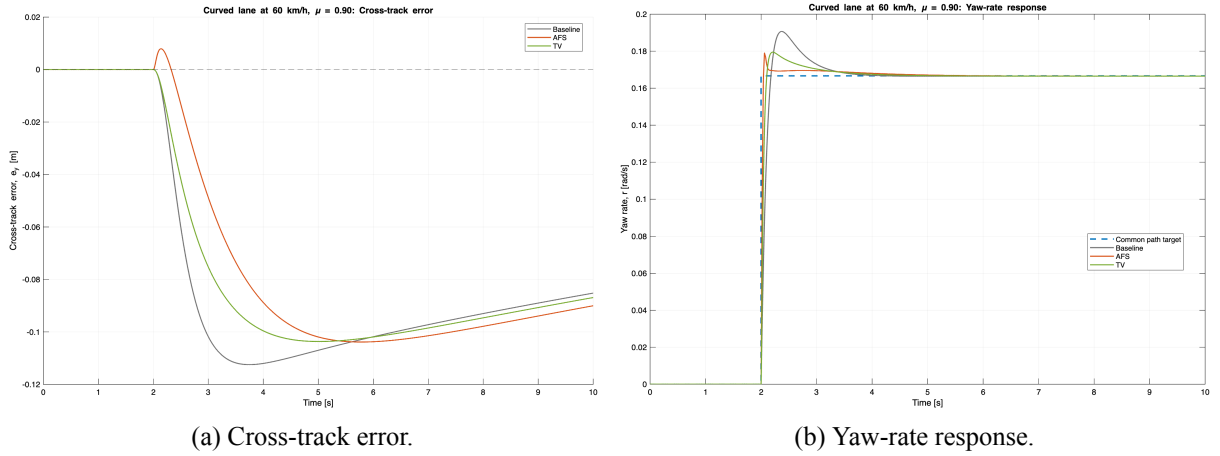


Figure 3: Bicycle-model response for the constant-radius curve.

The Baseline cross-track RMS error over the evaluation window is approximately 0.0999 m. AFS reduces this value by about 4.7%, while TV reduces it by about 3.4%. The improvement is therefore fairly small for both active controllers, although AFS gives the slightly lower path error.

The yaw-rate result shows a somewhat different trend. Compared with Baseline, the yaw-rate RMS tracking error is reduced by about 12.7% with AFS and by about 19.7% with TV. Therefore, TV gives the better yaw-rate tracking in this case even though AFS gives the slightly lower cross-track error. This difference is useful because it shows that a controller should not be ranked using only one quantity. Better yaw-rate tracking does not necessarily give the smallest path error, and the same is true in the opposite direction.

The required AFS correction is very small in this simulation because the Baseline bicycle model already follows the constant-radius path reasonably well. The correction stays well below the configured $\pm 5^\circ$ limit. The required TV yaw moment is also small compared with the ± 5000 N m limit. The sideslip responses are shown in Fig. 4.

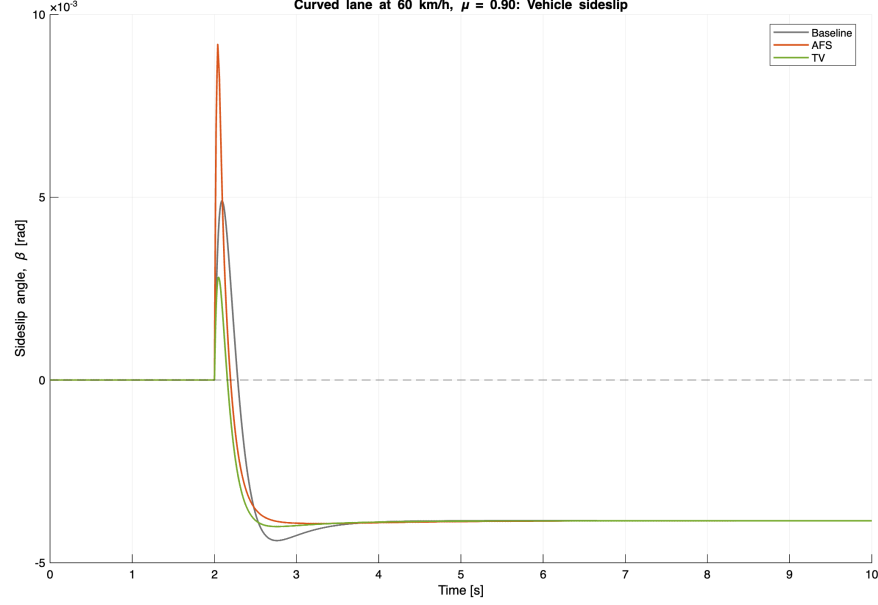


Figure 4: Bicycle-model sideslip response for the constant-radius curve.

The three sideslip responses remain close to one another. Overall, the bicycle model suggests that the Baseline controller already performs well for this moderate steady-cornering case. AFS and TV both improve the response, but the differences among the three cases remain relatively small.

D. Constant-Radius Curve: CarSim Tracking Results

A much clearer difference between the controllers is seen when the same maneuver is simulated using CarSim. Figure 5 shows the vehicle trajectory and cross-track error. The cross-track RMS values are summarized in Table 7.

Table 7: CarSim curved-lane cross-track performance.

Controller	Cross-track RMSE	Reduction from Baseline
Baseline	0.354 m	—
AFS	0.107 m	69.7%
TV	0.066 m	81.5%

The Baseline vehicle runs noticeably inside the desired circular path and has an RMS cross-track error of 0.354 m. This is much larger than the error obtained using the bicycle model. With AFS, the vehicle stays considerably closer to the desired path. The RMS error decreases to 0.107 m, which is a reduction of 69.7% compared with Baseline. TV gives the smallest cross-track error in this test. Its RMS

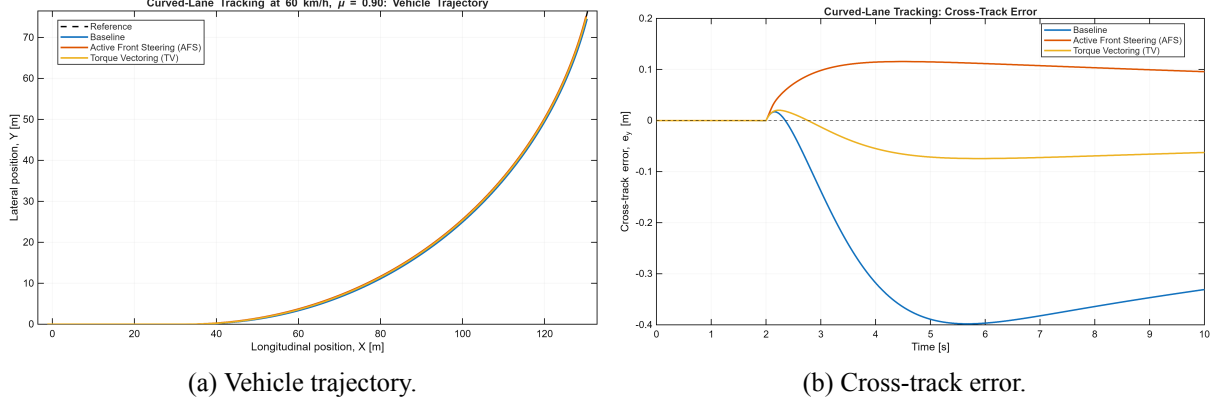


Figure 5: CarSim constant-radius curved-lane tracking at 60 km/h and $\mu = 0.90$.

value is 0.066 m, corresponding to an 81.5% reduction from Baseline. Based only on path tracking, TV therefore performs best for the constant-radius CarSim case.

The difference between the bicycle-model and CarSim results is also worth noting. In the bicycle model, the improvement from active control was only a few percent. In CarSim, both active controllers give much larger improvements. This suggests that the simplified model is useful for controller design and initial testing, but it does not capture all of the vehicle behavior that affects path tracking in the higher-fidelity model.

E. Constant-Radius Curve: Stability Response

The tracking results alone do not show the complete behavior of the vehicle. For this reason, the sideslip angle and lateral acceleration are also compared in Fig. 6.

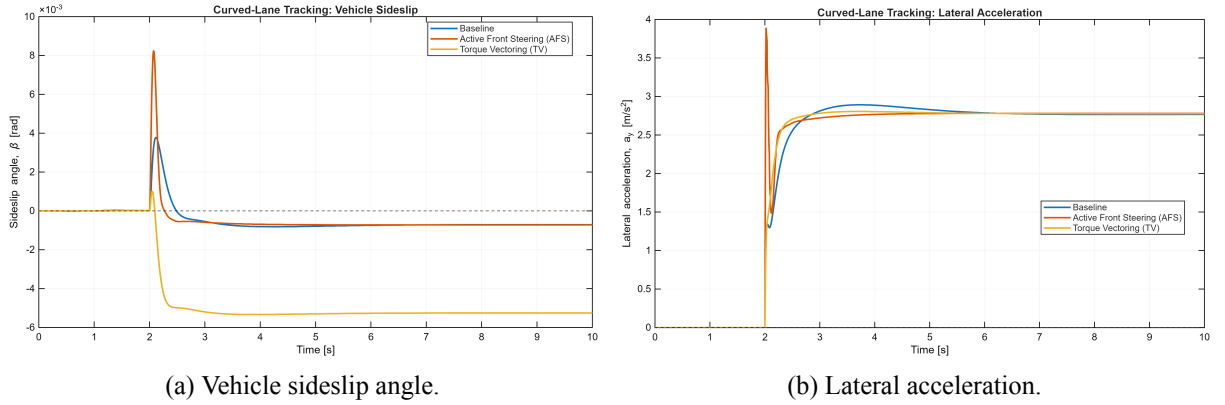


Figure 6: CarSim stability-related responses for the constant-radius curve.

After the initial curve-entry transient, the Baseline and AFS cases settle to relatively small sideslip angles. AFS therefore improves the path tracking considerably without producing a large change in steady sideslip. The TV case is different. Although TV gives the smallest cross-track error, its sideslip magnitude is noticeably larger. The peak value over the evaluation interval is approximately $|\beta|_{\text{peak}} \approx 0.306^\circ$. This value is still relatively small under the selected dry-road condition, so the result should not

be interpreted as an unstable vehicle. However, it does show that the best path tracking is obtained with a larger change in the direction of the vehicle velocity relative to the vehicle body.

The lateral acceleration of all three cases settles close to 2.78 m/s^2 , which is approximately the expected centripetal acceleration for a vehicle travelling at 60 km/h on a 100 m radius curve. Since all three vehicles are following approximately the same radius at the same speed, similar steady lateral acceleration is expected. Therefore, the main difference in this test is not the final lateral acceleration. It is how much steering correction or yaw moment is required to obtain the desired path, and how this affects sideslip.

F. Constant-Radius Curve: Actuator Effort

The control inputs for AFS and TV are shown in Fig. 7.

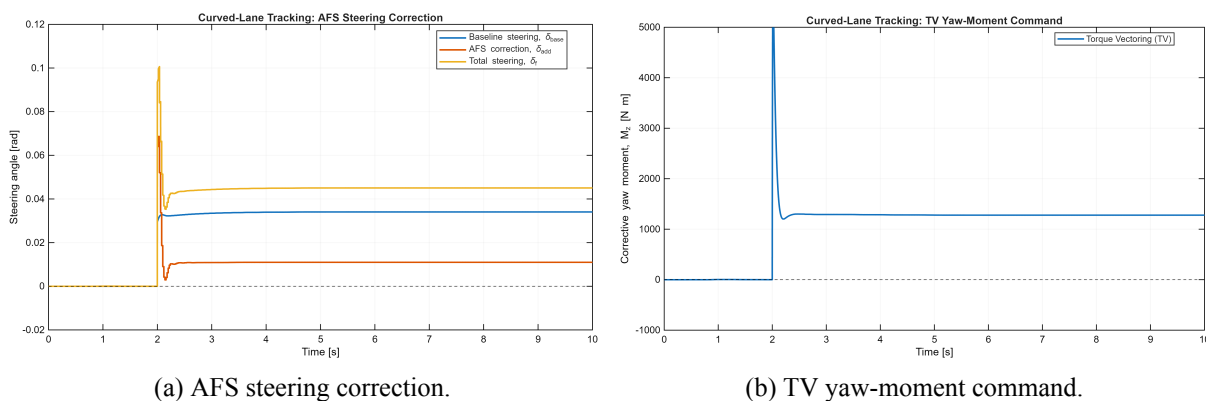


Figure 7: Active-control effort in the CarSim constant-radius curve.

AFS adds a steering correction when the vehicle enters the curve and then settles to a smaller correction once the response becomes steady. The additional steering stays within the $\pm 5^\circ$ AFS limit, and the total front steering remains below the $\pm 30^\circ$ limit. TV requires a much larger control action at curve entry. The requested yaw moment briefly reaches the configured 5000 N m limit and then settles to approximately 1250 N m during the steady part of the turn. This helps explain the strong tracking performance of TV. The direct yaw moment provides additional yaw authority without changing the front steering command. The cost is that a relatively large yaw-moment input is required, particularly during the curve-entry transient.

It should also be noted that the CarSim TV implementation in this project uses an equivalent external yaw moment. The controller does not allocate this moment to individual wheel torques. In a real torque-vectoring system, the required moment would have to be generated using left-right wheel torque differences while also considering motor limits, tire friction, and the available longitudinal tire forces.

G. Double Lane Change: Bicycle-Model Results

The double lane change is more demanding than the constant-radius curve because the steering and yaw direction change several times during the maneuver. The vehicle must move into the adjacent lane, settle briefly, and then return to the original lane.

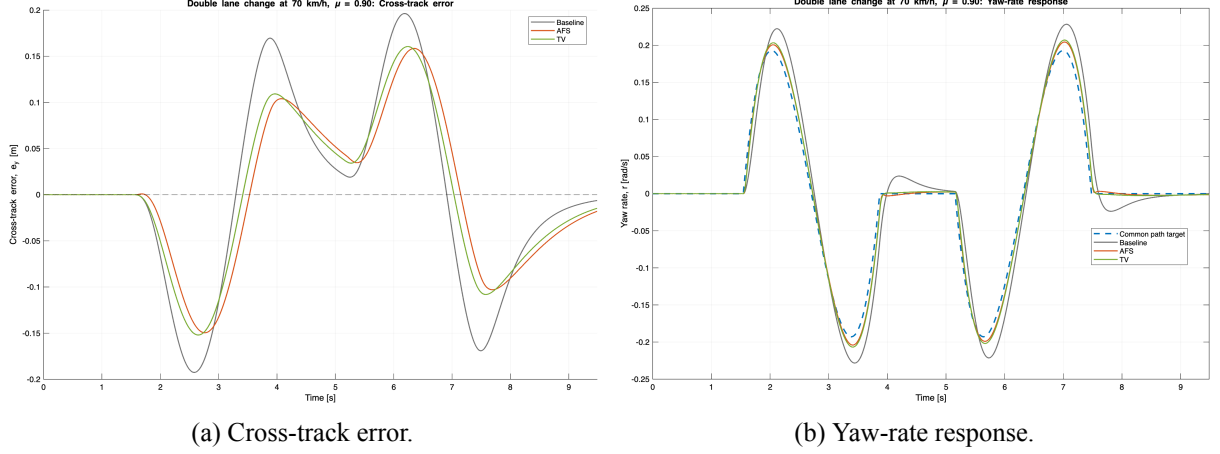


Figure 8: Bicycle-model response for the double lane change.

For this maneuver, both AFS and TV provide a more noticeable improvement than they did in the bicycle-model curved-lane test. Relative to Baseline, the cross-track RMS error is reduced by approximately 21.8% with AFS and 20.3% with TV. The yaw-rate improvement is larger. AFS reduces the yaw-rate RMS tracking error by approximately 67.2%, while TV reduces it by approximately 63.3%. The two active controllers therefore perform similarly, with AFS having a small advantage in both path tracking and yaw-rate tracking. The sideslip response is shown in Fig. 9.

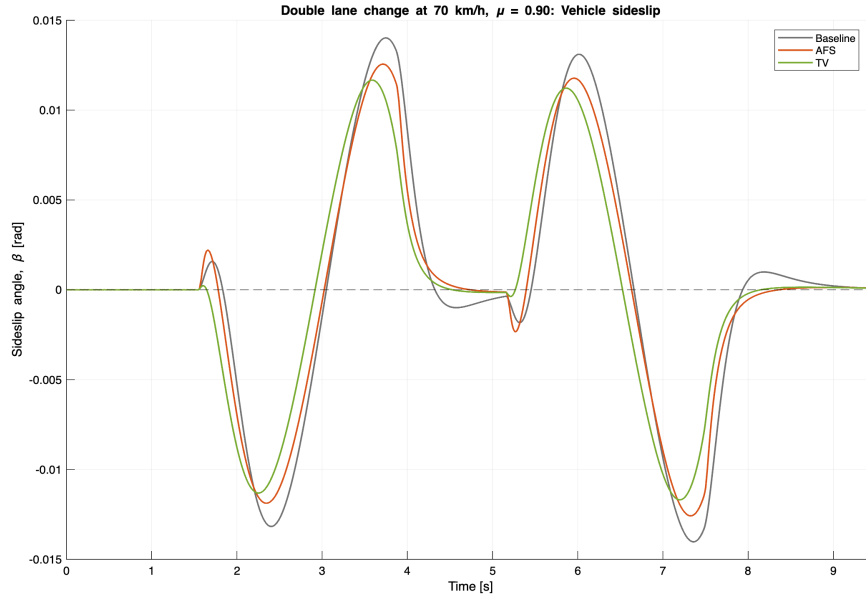


Figure 9: Bicycle-model sideslip response during the double lane change.

In the linear model, TV gives the smallest sideslip response, while AFS gives the slightly better tracking response. The difference between the two active controllers is still fairly small. Based on the bicycle model alone, both methods appear to be reasonable choices for the DLC.

H. Double Lane Change: CarSim Tracking Results

The higher-fidelity CarSim model gives a clearer separation between the controllers during the DLC. Figure 10 shows the trajectories and cross-track errors.

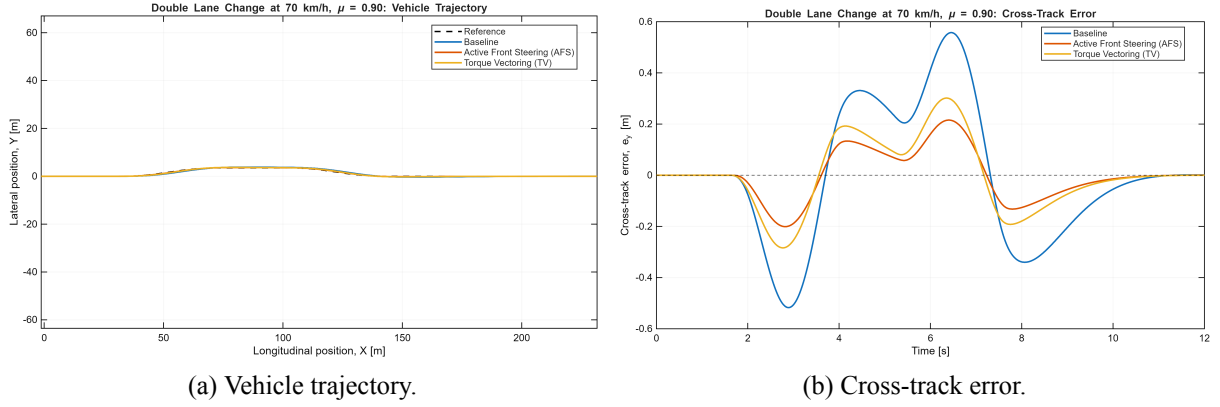


Figure 10: CarSim double-lane-change tracking at 70 km/h and $\mu = 0.90$.

The RMS tracking results are listed in Table 8.

Table 8: CarSim DLC cross-track performance.

Controller	Cross-track RMSE	Reduction from Baseline
Baseline	0.258 m	—
AFS	0.096 m	62.7%
TV	0.136 m	47.3%

The Baseline case shows the largest path deviation, especially during the two direction changes. The vehicle follows the general shape of the reference path, but its transient response produces noticeable overshoot. AFS gives the lowest cross-track error. The RMS error decreases from 0.258 m for Baseline to 0.096 m, which is a 62.7% improvement. TV also improves the response, reducing the RMS error to 0.136 m, or 47.3% below Baseline. However, it does not match the tracking accuracy obtained with AFS.

This result is different from the constant-radius CarSim case, where TV gave the lowest cross-track error. The difference suggests that the type of maneuver has a strong effect on which actuator performs better. One possible reason is the nature of the DLC itself. The maneuver requires rapid changes in the front steering direction. AFS acts directly through this same steering channel, so its correction can work together with the nominal path-following steering command. TV acts through a separate yaw moment and must reverse the corrective moment as the desired yaw direction changes.

I. Double Lane Change: Stability Response

The sideslip and lateral-acceleration results show another important difference between AFS and TV.

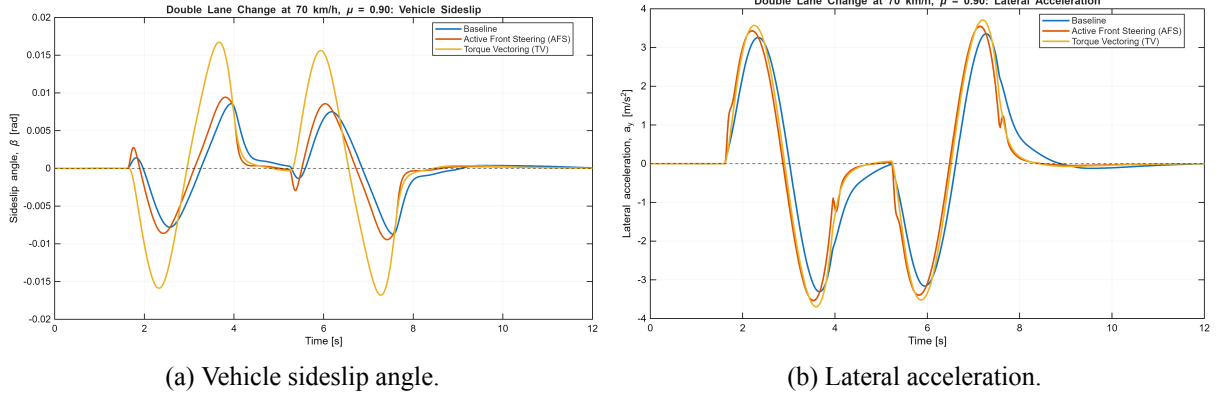


Figure 11: CarSim stability-related responses during the double lane change.

The peak sideslip values over the active maneuver are

$$|\beta|_{\text{peak}} = \begin{cases} 0.501^\circ, & \text{Baseline,} \\ 0.541^\circ, & \text{AFS,} \\ 0.963^\circ, & \text{TV.} \end{cases} \quad (68)$$

The AFS result is only slightly larger than the Baseline value. This is important because AFS gives the largest improvement in path tracking while keeping the peak sideslip relatively close to the uncontrolled case. TV produces a much larger peak sideslip of approximately 0.963° . The TV controller is still able to reduce the path error, but the vehicle develops more lateral motion relative to its body orientation while doing so. The TV case also produces the largest lateral-acceleration peak. This is consistent with the more aggressive yaw correction during the lane-change transients. For this maneuver, AFS therefore gives the better overall balance between path tracking and lateral stability. TV improves the path relative to Baseline, but the improvement comes with noticeably larger sideslip and lateral acceleration.

J. Double Lane Change: Actuator Effort

Figure 12 shows the control effort required by AFS and TV during the DLC.

The AFS correction follows the main steering changes throughout the maneuver. Since the MPC penalizes changes in the additional steering command, the correction remains smooth rather than changing abruptly. The AFS command stays within the $\pm 5^\circ$ correction limit, and the total front steering remains within the $\pm 30^\circ$ constraint. The TV yaw moment changes sign as the yaw direction reverses. Its peak magnitude is approximately 2.47 kN m. Unlike the curved-lane case, TV does not reach the ± 5 kN m limit during the main DLC maneuver. However, it still requires repeated positive and negative yaw-moment inputs. In a real wheel-level TV system, these moment changes would have to be produced by redistributing the longitudinal wheel forces and motor torques.

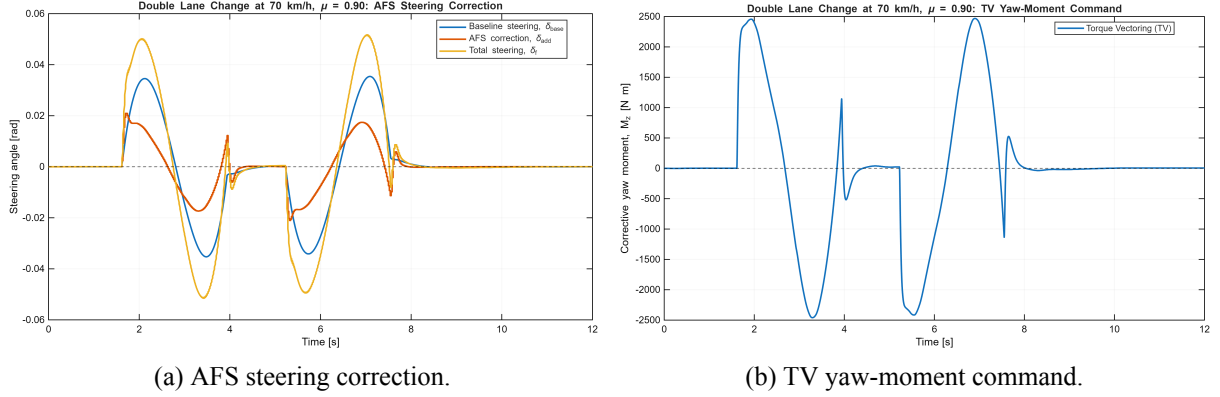


Figure 12: Active-control effort during the CarSim double lane change.

K. Effect of Vehicle-Model Fidelity

The comparison between the bicycle model and CarSim is one of the more interesting parts of the results. The controller rankings are not exactly the same when the vehicle model is changed. The bicycle model only includes the main planar lateral and yaw dynamics. This makes the equations simple enough for controller design and helps show clearly how steering and yaw moment affect β and r . For that reason, it was useful for developing the AFS prediction model, deriving the TV feedforward contribution, and checking the controller implementation. At the same time, several effects are not included in the bicycle model. These include nonlinear tire behavior, load transfer, suspension motion, steering geometry and compliance, combined longitudinal and lateral tire forces, and individual wheel dynamics. CarSim includes a more detailed representation of these effects.

The difference is most obvious in the curved-lane tracking results. In the bicycle model, AFS and TV reduce the cross-track RMS error by only 4.7% and 3.4%, respectively. In CarSim, the reductions are 69.7% for AFS and 81.5% for TV. The DLC shows a similar change. The bicycle model predicts very similar tracking performance for AFS and TV, while CarSim gives a clearer difference. AFS reduces the cross-track RMS error by 62.7%, compared with 47.3% for TV. CarSim also predicts a much larger sideslip response for TV than the bicycle model does.

This does not mean that the bicycle model is unsuitable. It serves a different purpose. The linear model is useful for understanding the basic vehicle dynamics and designing the controllers. The CarSim results show, however, that final performance and controller selection should also be checked using a higher-fidelity vehicle model.

L. Overall Comparison

Table 9 summarizes the main observations from the four sets of results.

The results do not point to one controller being better in every situation. Instead, the better choice depends on the maneuver and on which performance measure is given the highest priority.

For the constant-radius CarSim test, TV gives the best path tracking. However, it also produces the largest sideslip and requires a large yaw moment at curve entry. AFS does not reduce the cross-track

Table 9: Overall qualitative controller comparison.

Case	Tracking result	Sideslip behavior	Actuator demand	Interpretation
Bicycle curve	AFS and TV only slightly better than Baseline	Nearly identical	Low	No clear overall winner; differences are small in the linear model.
CarSim curve	TV best; AFS second	TV higher than AFS/Baseline	TV reaches 5 kN m limit	TV gives the lowest path error, while AFS keeps sideslip closer to Baseline.
Bicycle DLC	AFS slightly best tracking; TV similar	TV gives lowest linear-model sideslip	Moderate	Both active controllers improve the response, with only small differences.
CarSim DLC	AFS best; TV second	TV largest, approximately 0.963° peak	TV reaches about ± 2.47 kN m	AFS gives the best overall balance of tracking and stability.

error as much as TV, but it still gives a major improvement compared with Baseline while keeping the sideslip response much closer to the original vehicle. For the CarSim double lane change, AFS performs better overall. It gives the lowest cross-track error and only a small increase in peak sideslip relative to Baseline. TV still improves the path tracking, but the larger sideslip and lateral acceleration make the result less balanced.

The comparison therefore suggests that actuator choice should depend on the maneuver and the control objective. TV is very effective when strong direct yaw correction is needed, while AFS works particularly well in the tested transient lane-change maneuver where the main vehicle motion is already being created through steering.

M. Limitations of the Present Study

There are several limitations to the present comparison. First, the TV controller is applied in CarSim as an equivalent external yaw moment. The simulation does not calculate the individual wheel torques needed to generate that moment. In a real electric vehicle, the achievable yaw moment would depend on motor limits, track width, normal loads, available longitudinal tire force, and the tire friction limit. Therefore, the TV results in this project represent the vehicle-level yaw-moment controller rather than a complete wheel-level torque-vectoring system.

Second, the AFS controller continues to use a linear bicycle model for prediction when it is connected to the CarSim vehicle. This means that the optimizer does not directly predict nonlinear tire saturation, load transfer, suspension effects, or detailed steering dynamics. The feedback loop helps correct for this model mismatch at each controller update, but the prediction itself remains simplified. Third, only

two maneuvers and one road-friction condition, $\mu = 0.90$, are included in the final comparison. These results therefore show how the controllers behave for the selected curved-lane and double-lane-change tests, but they do not cover the complete vehicle operating range. Lower-friction conditions or higher speeds could change the relative performance of AFS and TV.

The controller tuning should also be kept in mind when comparing the results. The AFS weights and constraints were selected to give stable constrained performance, while the TV feedback structure retains the literature-based PID gains used in the project. The results therefore compare the implemented versions of the two controllers and should not be taken as the absolute best performance possible from either AFS or TV. Finally, other practical factors such as passenger comfort, electrical energy consumption, tire wear, actuator temperature, and implementation cost were not included. These factors would also be important when choosing between AFS and TV for a production vehicle.

V. Conclusions and Future Work

This project compared Active Front Steering (AFS) and Torque Vectoring (TV) as two different ways of improving path tracking and yaw stability. Both controllers were added to the same baseline path-following system so that the comparison could be made under the same vehicle parameters, maneuvers, road conditions, and evaluation metrics. AFS modified the front steering angle, while TV acted through a corrective yaw moment.

The linear bicycle model was useful during the first stage of the project because it provided a simple way to understand how steering, sideslip, yaw rate, and yaw moment are related. It was also used directly in the AFS prediction model and in the TV feedforward calculation. However, the results showed that conclusions from the linear model alone were not enough to judge the final controller performance.

This became particularly clear when the controllers were tested in CarSim. For the constant-radius curve, the bicycle model predicted only small differences between Baseline, AFS, and TV. In CarSim, the difference was much larger. Both active controllers improved the tracking significantly, with TV giving the lowest cross-track error. However, TV also produced a larger sideslip response and required a large corrective yaw moment, including saturation during curve entry. AFS did not achieve the same minimum tracking error, but its sideslip response remained much closer to the Baseline vehicle.

The double lane change gave a different result. In this maneuver, AFS provided the best overall performance in CarSim. It gave the lowest cross-track error while keeping the peak sideslip only slightly above the Baseline case. TV still improved the tracking compared with Baseline, but its sideslip and lateral acceleration were noticeably larger. The difference between these two maneuvers was one of the main findings of the project. TV was very effective in the steady curved-lane case, where a direct yaw moment could continuously help the vehicle hold the desired path. The double lane change involved repeated steering and yaw reversals. In this case, AFS benefited from acting through the same front steering channel that was already being used to perform the maneuver. This allowed the corrective steering to work together with the nominal path-following command. Another important observation was the effect of model fidelity. The bicycle model was valuable for controller design and for understanding the main vehicle dynamics, but it did not reproduce the full differences seen in CarSim. The CarSim

model includes effects that are not represented in the linear model, such as more detailed tire behavior, load transfer, steering geometry, suspension effects, and individual wheel dynamics. For this reason, the project showed the importance of validating a controller using a higher-fidelity vehicle model before making a final comparison.

Overall, neither AFS nor TV was better in every situation. TV gave the best tracking in the constant-radius CarSim test, while AFS gave the best balance of tracking and stability in the CarSim double lane change. The choice between the two actuators therefore depends on the maneuver and on what is being prioritized. If the main objective is strong direct yaw correction, TV can be very effective. If the maneuver is mainly steering-driven and a smaller sideslip response is important, AFS may be the better choice.

There are several directions in which this work could be extended. The first is to combine AFS and TV rather than using them separately. The results suggest that the two actuators have complementary strengths. A coordinated controller could use AFS for normal steering correction and use TV when additional yaw authority is needed. The controller would then have to decide how much correction should be assigned to steering and how much should be assigned to the yaw-moment actuator.

A second improvement would be to replace the equivalent external yaw moment used in the current CarSim TV implementation with an actual wheel-level torque allocator. In the present project, CarSim receives the desired yaw moment directly. A real torque-vectoring system would need to convert this moment into individual left- and right-wheel motor torques. This would make it possible to include motor limits, wheel normal loads, combined tire-force limits, regenerative braking, and longitudinal torque demand in the analysis. The operating conditions should also be expanded. The final simulations were performed at $\mu = 0.90$, so future work should include lower-friction roads, different vehicle speeds, and more aggressive paths. These cases would be especially useful for studying when the available tire force becomes a limiting factor for AFS or TV.

The AFS controller could also be extended to use a nonlinear prediction model. The current controller uses a speed-dependent linear bicycle model, which keeps the optimization relatively simple. A nonlinear MPC formulation could include nonlinear tire forces and explicit tire-force or sideslip constraints, although this would increase the computational requirements. Other performance measures could also be included in future comparisons. For example, lateral jerk and yaw acceleration could be used to evaluate passenger comfort. Energy consumption and motor usage would be important for TV, while steering activity and steering actuator effort could be examined for AFS. Tire wear and thermal limits would also become important in a more complete vehicle-level study.

Finally, the controllers should eventually be tested beyond software simulation. Hardware-in-the-loop testing would be a useful next step because it would introduce real-time computation and actuator delays while still allowing the vehicle to be simulated safely. Experimental vehicle testing could then be used to study the effects of sensor noise, parameter uncertainty, friction variation, and other dynamics that are difficult to capture completely in simulation.

In summary, both AFS and TV were able to improve the response of the vehicle, but they did so in different ways. The results showed that good path tracking should not be considered separately from

sideslip, lateral acceleration, and actuator effort. They also showed that the preferred controller can change with the maneuver and with the level of detail in the vehicle model. For this reason, selecting a yaw-stability actuator should be based on the overall vehicle response rather than on a single tracking or stability metric.

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